CHF / Acute Pulmonary Edema

AC-05 | Adult Cardiac

Quick flow

01 Suspected Pulmonary Edema [EMT / AEMT / Paramedic]

Acute dyspnea • orthopnea • rales • diaphoresis • hypertension • JVD/edema • frothy sputum

02 Position + Monitor [EMT / AEMT / Paramedic]

Upright unless hypotensive • SpO_2 • frequent BP • cardiac monitor • IV • 12-lead without delaying support

03 Moderate / Severe Distress? [EMT / AEMT / Paramedic]

Spontaneously breathing • airway protected • cooperative enough for mask • SBP ≥100

04 Start CPAP Early [EMT / AEMT / Paramedic]

If adjustable: start 5 cm H_2O and titrate to 10 • fixed device: manufacturer setting • reassess continuously

05 Nitroglycerin Eligible? [EMT / AEMT / Paramedic]

SBP ≥110 before every dose • no shock, RV infarction/preload dependence, or prohibited PDE-5 use

06 Nitroglycerin [EMT / AEMT / Paramedic]

0.4 mg SL every 5 min • maximum 3 • EMT assists prescribed medication • AEMT/Paramedic agency medication

07 Shock or Ventilatory Failure? [EMT / AEMT / Paramedic]

Stop nitrates • reduce/remove CPAP if worsening hypotension • BVM support and request Paramedic airway/shock care

08 Paramedic - Cardiogenic Shock [Paramedic]

Norepinephrine per the AC-05 dose and perfusion targets below; continuous cardiac monitoring and frequent blood-pressure and vascular-site checks

09 Reassess + Transport [EMT / AEMT / Paramedic]

BP • work of breathing • mental status • SpO_2/$EtCO_2$ • lung sounds • early alert • scene target <15 min

Overview

- Adult pathway for patients 16 years of age and older with suspected cardiogenic pulmonary edema or acute decompensated heart failure.
- Prioritize early noninvasive positive-pressure ventilation and blood-pressure-guided vasodilation while identifying acute coronary syndrome, dysrhythmia, cardiogenic shock, and alternative causes of respiratory distress.
- Target scene time is less than 15 minutes unless essential stabilization requires additional time.

EMT Care

Rapid recognition, positioning, oxygenation, CPAP, prescribed-nitroglycerin assistance, monitoring, and early transport.

EMT

- Perform the primary assessment; position upright unless hypotensive and assess respiratory effort, lung sounds, perfusion, mental status, SpO_2, and complete vital signs.
- Apply the cardiac monitor/AED and obtain a 12-lead ECG when trained and equipped, without delaying respiratory support.
- Start CPAP early for moderate-to-severe distress when the patient is breathing spontaneously, can protect the airway, can tolerate the mask, and has SBP of at least 100 mmHg.

- Titrate oxygen to SpO_2 92-96%; use 88-92% for known chronic hypercapnic respiratory failure unless severe hypoxemia requires a higher target.
- Assist with the patient's prescribed nitroglycerin only when SBP is at least 110 mmHg and no contraindication is present.
- Prepare for BVM ventilation, rapid transport, and ALS support if mental status, ventilation, perfusion, or CPAP tolerance worsens.

AEMT Care

All EMT care plus vascular access, agency nitroglycerin, CPAP management, and recognition of cardiogenic shock.

AEMT

- Perform all EMT actions and establish IV access without delaying CPAP, nitroglycerin, or transport.
- Give nitroglycerin 0.4 mg SL every 5 minutes, maximum three doses, only when SBP is at least 110 mmHg before every dose and no contraindication is present.
- Reassess blood pressure, symptoms, work of breathing, mental status, SpO_2, and perfusion after every nitroglycerin dose and CPAP adjustment.
- Do not administer a routine fluid bolus when pulmonary edema is present.
- Support ventilation with BVM and immediately request paramedic intervention when shock, ventilatory failure, or inability to protect the airway develops.

Paramedic Care

All AEMT care plus advanced ECG assessment, cardiogenic-shock treatment, advanced airway management, and linked cause-specific care.

Paramedic

- Perform all EMT and AEMT actions; interpret the 12-lead ECG and evaluate for ACS, dysrhythmia, acute valvular disease, hypertensive pulmonary edema, and cardiogenic shock.
- For persistent cardiogenic shock, start norepinephrine at 0.1 mcg/kg/min IV/IO and titrate to MAP of at least 65 mmHg or SBP of at least 90 mmHg; maximum 2 mcg/kg/min.
- Withhold nitroglycerin and reduce or discontinue CPAP when either intervention is worsening hypotension or perfusion.
- Use waveform $EtCO_2$ when available and technically compatible with CPAP; trend ventilation and response rather than relying on a single value.
- Prepare for drug-assisted airway management when CPAP fails, ventilation is inadequate, mental status declines, or the airway cannot be protected; anticipate peri-intubation hypotension.
- Open AC-04 for suspected ACS/STEMI, AM-05 for shock, AR-03 for drug-assisted airway management, or AC-03 for cardiac arrest.

Indications

- Suspected cardiogenic pulmonary edema or acute decompensated heart failure in a patient 16 years of age or older.
- Acute dyspnea with orthopnea, rales, diaphoresis, hypertension, jugular venous distention, peripheral edema, frothy sputum, or a history strongly suggestive of heart failure.
- Respiratory distress associated with suspected acute cardiac ischemia, dysrhythmia, valvular disease, renal failure, or medication nonadherence when pulmonary edema is clinically suspected.

Contraindications

- CPAP is contraindicated for apnea, inadequate spontaneous ventilation, active vomiting, inability to protect the airway, significant facial trauma preventing a seal, suspected untreated pneumothorax, severe agitation preventing safe application, or SBP below 100 mmHg.

- Nitroglycerin is contraindicated when SBP is below 110 mmHg, shock or preload dependence is present, right-ventricular infarction is suspected, or prohibited recent PDE-5 inhibitor use is identified.
- Do not administer routine large-volume crystalloid when pulmonary edema is present.

Assessment | Focused history

- Determine onset and progression, orthopnea, paroxysmal nocturnal dyspnea, chest discomfort, palpitations, syncope, fever, cough, sputum, medication adherence, dialysis schedule, recent weight gain, and baseline oxygen or CPAP use.
- Identify prior heart failure, myocardial infarction, PCI/CABG, valvular disease, renal failure, dysrhythmia, COPD/asthma, pulmonary embolism, and implanted cardiac devices.
- Review nitrates already taken, diuretics and antihypertensives, antiplatelet or anticoagulant therapy, and recent PDE-5 inhibitor use.

Assessment | Examination and monitoring

- Assess respiratory rate and pattern, accessory-muscle use, speech, lung sounds, diaphoresis, cyanosis, frothy sputum, JVD, edema, pulse quality, capillary refill, and mental status.
- Obtain continuous cardiac monitoring, SpO_2, frequent blood pressures, and two complete vital-sign sets when feasible.
- Obtain a 12-lead ECG without delaying CPAP, ventilation, or transport; use AC-04 when ACS or STEMI is suspected.
- Use waveform $EtCO_2$ when available and technically compatible with CPAP or assisted ventilation.

Assessment | Alternative and precipitating diagnoses

- Consider acute coronary syndrome, hypertensive emergency, dysrhythmia, acute valvular disease, renal failure, medication nonadherence, infection, and dietary or fluid excess as precipitants.
- Consider COPD/asthma, pneumonia, pulmonary embolism, pneumothorax, aspiration, anaphylaxis, metabolic acidosis, and other causes of respiratory distress.
- Wheezing may occur with pulmonary edema; do not assume asthma or COPD solely because wheezing is present.

Treatment steps

- Position upright unless hypotensive and begin continuous cardiac monitoring, SpO_2, frequent blood-pressure measurement, and focused reassessment.
- Titrate oxygen to SpO_2 92-96%; use 88-92% for known chronic hypercapnic respiratory failure unless severe hypoxemia requires a higher target.
- Start CPAP early for moderate-to-severe respiratory distress when the patient is breathing spontaneously, can protect the airway, can tolerate the mask, and has SBP of at least 100 mmHg.
- For adjustable CPAP, begin at 5 cm H_2O and titrate to 10 cm H_2O according to response and tolerance. For a fixed-pressure device, use the manufacturer-approved setting.
- Assist with prescribed nitroglycerin at the EMT level or administer agency nitroglycerin at the AEMT/Paramedic level: 0.4 mg SL every 5 minutes as needed, maximum three doses.
- Confirm SBP of at least 110 mmHg before every nitroglycerin dose. Withhold for shock, suspected RV infarction or preload dependence, avanafil within 12 hours, sildenafil or vardenafil within 24 hours, or tadalafil within 48 hours.
- Reassess blood pressure, respiratory effort, mental status, SpO_2, lung sounds, perfusion, and mask tolerance after every nitroglycerin dose and CPAP adjustment.
- Do not routinely administer furosemide, another diuretic, morphine, another opioid, or an IV fluid bolus under this protocol.

- Administer albuterol only when true bronchospasm is suspected; wheezing alone may be caused by pulmonary edema.
- For hypotension or cardiogenic shock, withhold nitroglycerin, reduce or discontinue CPAP if it worsens perfusion, support ventilation, and avoid routine crystalloid when pulmonary edema is present.
- For persistent cardiogenic shock, a paramedic may start norepinephrine at 0.1 mcg/kg/min IV/IO and titrate to MAP of at least 65 mmHg or SBP of at least 90 mmHg; maximum 2 mcg/kg/min.
- Prepare for advanced airway management when CPAP fails, ventilation is inadequate, mental status declines, or the airway cannot be protected.
- Notify the receiving facility early, treat while moving toward transport, and target scene time under 15 minutes.

Medications

Nitroglycerin

0.4 mg SL every 5 minutes as needed; maximum 3 doses

- EMT may assist with the patient's prescribed nitroglycerin; AEMT and Paramedic may administer agency nitroglycerin.
- Confirm SBP of at least 110 mmHg before every dose.
- Withhold for shock, suspected right-ventricular infarction or preload dependence, avanafil within 12 hours, sildenafil or vardenafil within 24 hours, or tadalafil within 48 hours.
- IV bolus, infusion, and other high-dose nitroglycerin regimens are not authorized under AC-05.

Norepinephrine

Start 0.1 mcg/kg/min IV/IO; titrate to MAP ≥65 mmHg or SBP ≥90 mmHg; maximum 2 mcg/kg/min

- Paramedic standing order for persistent cardiogenic shock.
- Continuous cardiac monitoring, frequent blood-pressure assessment, and careful IV/IO-site surveillance are required.
- Use a proximal, well-functioning IV or IO and reassess perfusion frequently.

Albuterol

Use the approved respiratory dose only when true bronchospasm is suspected

- Do not administer solely because wheezing is present; pulmonary edema may produce cardiac wheezing.

Warnings / precautions

- Do not delay CPAP, ventilation, or transport for IV access or a 12-lead ECG.
- Do not use CPAP for apnea, inadequate spontaneous ventilation, inability to protect the airway, active vomiting, or shock.
- Positive-pressure ventilation can worsen hypotension; reassess blood pressure and perfusion continuously.
- Withhold nitroglycerin for SBP below 110 mmHg, shock, suspected right-ventricular infarction or preload dependence, and prohibited recent PDE-5 inhibitor use.
- Do not routinely administer a diuretic, opioid, or IV fluid bolus under AC-05.
- Norepinephrine requires continuous cardiac monitoring, frequent blood-pressure measurement, and IV/IO-site assessment.
- Intubation in cardiogenic shock may precipitate profound hypotension or cardiac arrest; optimize oxygenation and hemodynamics and prepare vasopressor support.
- Use CPAP only with confirmed equipment capabilities and device-specific training. Follow the fixed or adjustable device instructions and the pressure limits specified here.

Clinical pearls

- Early CPAP and blood-pressure-guided nitroglycerin are the principal field interventions for hypertensive cardiogenic pulmonary edema.
- Acute pulmonary edema may reflect rapid fluid redistribution rather than total-body fluid overload; a single physical finding does not establish the diagnosis.
- Wheezing does not exclude pulmonary edema, and rales do not exclude COPD, pneumonia, or another respiratory diagnosis.
- A normal initial ECG does not exclude acute coronary syndrome; use AC-04 when the history or tracing is concerning.
- Improvement in speech, respiratory rate, work of breathing, mental status, SpO_2, and blood pressure provides a useful treatment-response trend.

Special populations | Renal failure or dialysis

- Determine the last completed dialysis treatment, missed treatments, access complications, baseline weight, and known potassium problems.
- Avoid routine fluid administration and consider hyperkalemia when bradycardia, weakness, peaked T waves, or QRS widening is present.

Special populations | Cardiogenic shock

- Hypotension, altered mental status, cool or mottled skin, weak pulses, delayed capillary refill, and reduced urine output suggest poor perfusion.
- Withhold nitrates, limit positive pressure when it worsens perfusion, avoid routine crystalloid in pulmonary edema, and start norepinephrine when indicated.

Related pathways / contacts

AC-04 ACS / STEMI - Open for suspected ischemia, STEMI, or acute coronary occlusion.

AM-05 Hypotension / Shock - Open for persistent hypotension or shock management.

AR-03 Drug-Assisted Airway - Open when CPAP fails or airway protection is lost.

AC-03 Adult Cardiac Arrest - Open immediately if the patient becomes pulseless.

Source and references

American Heart Association/American College of Cardiology/Heart Failure Society of America. 2022 Guideline for the Management of Heart Failure. | Tennessee Emergency Medical Services Board. Tennessee EMS Protocol Guidelines, September 2025. | Claiborne EMS medication formulary and related AC-03, AC-04, AM-05, and AR-03 protocols.

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